Phishing Email Investigation & Quarantine Validation

Microsoft Defender XDR | Microsoft 365 | Authorized SOC Lab

Investigation date: 14 October 2026 • Exercise: Microsoft SOC Analyst Challenge

Executive summary

I conducted an authorized phishing simulation using a refund-themed email sent to a Microsoft 365 test mailbox. Microsoft Defender initially classified the message as phishing and quarantined it before normal inbox delivery.

I investigated the message using Microsoft Defender XDR, reviewing sender authentication, detection technologies, embedded URLs, and delivery actions. I then used KQL Advanced Hunting to independently verify the detection and correlate the original email with a recorded URL-click event.

During the controlled exercise, an authorized administrator released the message to the recipient's inbox. Defender subsequently recorded an allowed click on the refund-verification URL. The evidence confirmed successful initial quarantine and administrative release but did not establish credential theft, malware execution, or account compromise.

Outcome	**Initially blocked and quarantined; authorized administrative release succeeded; one allowed URL-click event recorded.**
Detection policy	Ronap-Lab-Anti-PhishingPolicy
Detection signals	User impersonation and advanced filtering
Severity / risk	Simulated phishing; no confirmed user compromise
Scope	Authorized test accounts and controlled lab email

Objective and scope

Validate Microsoft 365 anti-phishing protections against a controlled refund-themed phishing simulation and document the investigation from initial detection through final disposition.

The investigation included email authentication analysis, detection-policy validation, KQL threat hunting, quarantine release verification, and URL-click telemetry analysis.

The exercise did not involve credential theft or malware execution. Although an allowed URL click was recorded, the available evidence did not establish malicious URL blocking or account compromise.

Attack simulation

Sender	**phishing-sender@example.com**
Recipient	test-recipient@example.com
Subject	Action Required: Your $485.75 Refund Is Pending

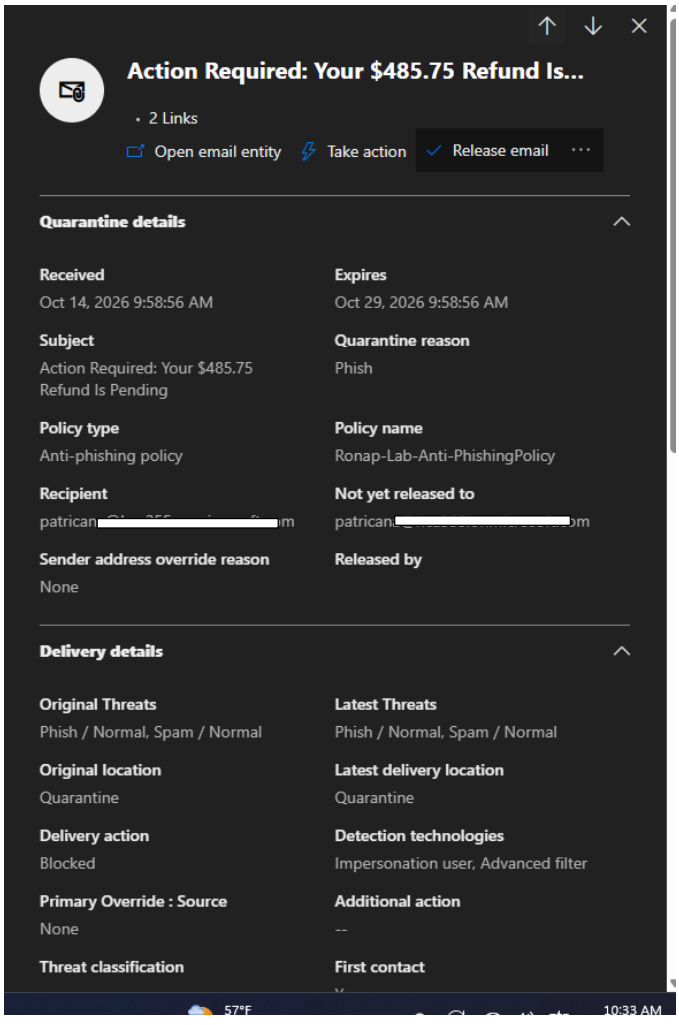
Social engineering	Financial incentive, urgency, and request to verify a refund
Embedded URLs	https://example.com/refund-verification • http://www.link.com/
Attachments	None

The message was intentionally constructed to resemble a plausible refund notice. The links were included for controlled analysis; the available evidence does not establish either URL as malicious.

Detection and triage evidence

The initial quarantine record classified the message as Phish and identified the configured anti-phishing policy. The original delivery action was Blocked, with Quarantine as the original delivery location. Detection technologies included "Impersonation user" and "Advanced filter."

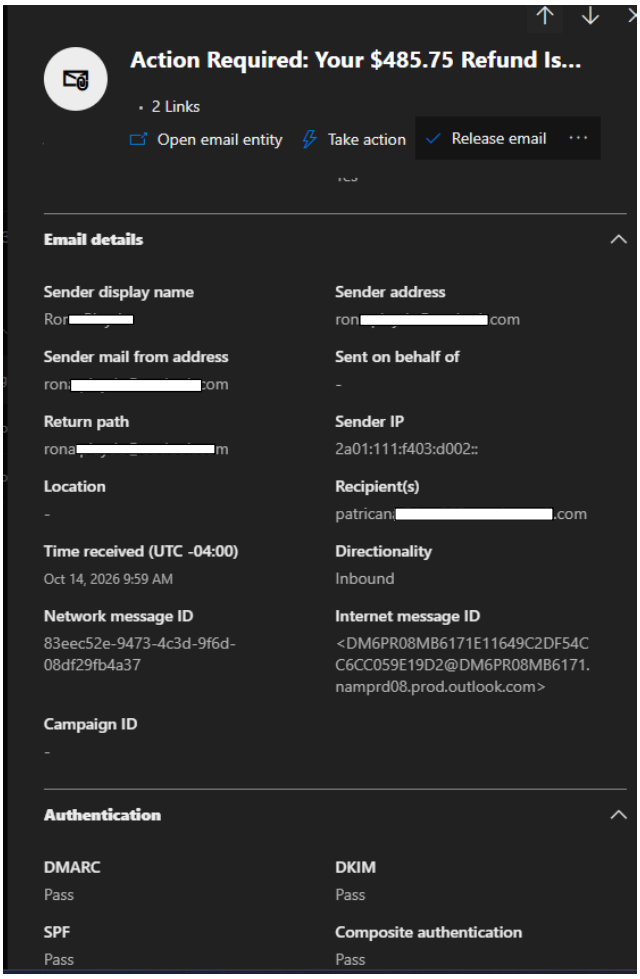
A later quarantine review confirmed that an authorized administrator successfully released the message, changing its latest delivery location to Inbox/folder. The initial quarantine decision remained valid despite the subsequent administrative release.



Evidence 01 — Quarantine details, detection policy, and blocked delivery.

Sender and authentication review

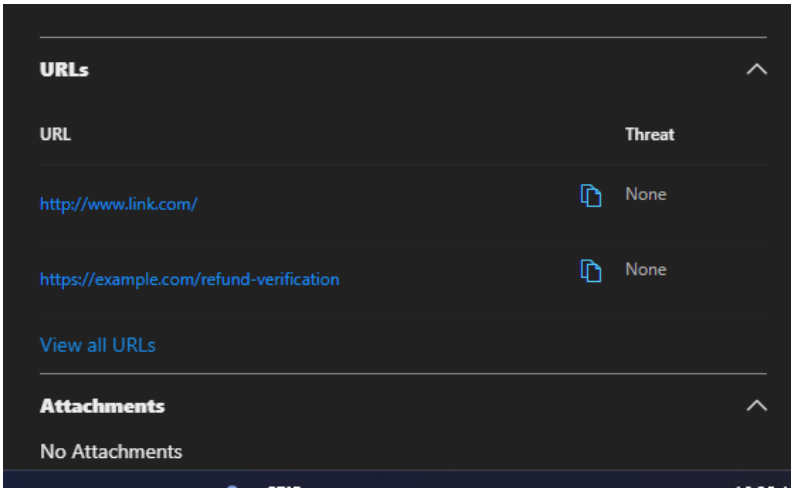
The email entity lists the external sender and the intended Microsoft 365 recipient. SPF, DKIM, DMARC, and composite authentication are shown as Pass. These authentication passes do not make an email safe; authenticated messages can still use deceptive content or impersonation patterns.



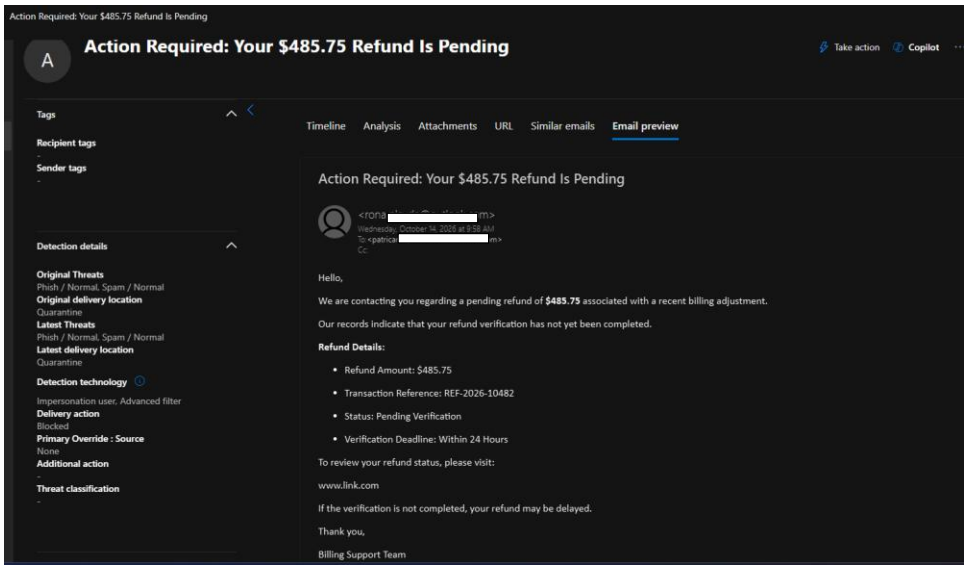
Evidence 02 — Message identity, routing information, and authentication results.

URL and message-content analysis

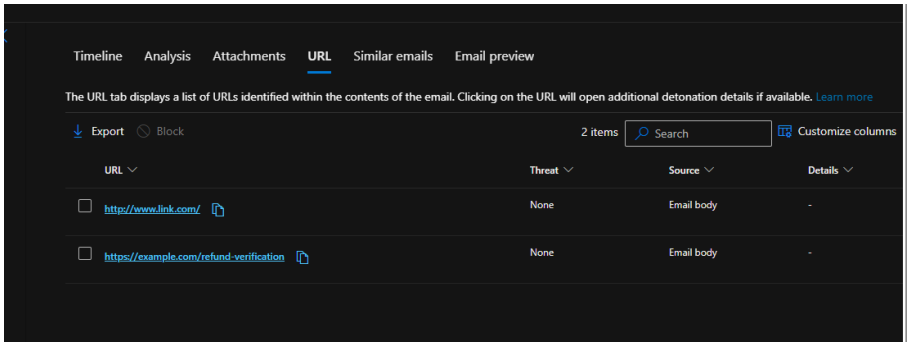
The email entity extracted two URLs from the message body. Both showed a threat verdict of “None” in the captured view. The email was nevertheless quarantined because of phishing-related detection, not because the screenshots demonstrate a Safe Links malicious-URL block.



Evidence 03 — Extracted URLs and absence of attachments.



Evidence 04 — Refund-themed message preview and associated phishing indicators.



Evidence 05 — URL tab listing both extracted links with verdict “None.”

SOC investigation workflow

1. Submitted a controlled refund-themed message from the external test mailbox to the Microsoft 365 lab recipient.

1. 2. Reviewed Microsoft Defender quarantine to confirm delivery disposition and the policy responsible.
2. Opened the email entity and examined sender, recipient, subject, and authentication results.
3. Inspected the email preview for financial urgency and the embedded verification request.
4. Reviewed extracted URLs, threat verdicts, and attachment status.
6. Recorded the supported conclusion: anti-phishing protection quarantined the message; Safe Links blocking was not demonstrated.

5. Queried the EmailEvents table in Microsoft Defender XDR Advanced Hunting using the original message's NetworkMessageId to independently validate the phishing classification, detection methods, and original delivery action.

6. Queried UrlClickEvents using the same NetworkMessageId and identified one ClickAllowed event associated with the refund-verification URL.

7. Reviewed the original message's quarantine details and confirmed that an authorized lab administrator successfully released it to the intended recipient.

8. Correlated the original detection, quarantine release, and URL-click telemetry to establish the final investigation outcome and document the limitations of the available evidence.

Findings and analyst assessment

Confirmed	**Message was categorized as Phish and blocked into Quarantine.**
Confirmed	Anti-phishing policy Ronap-Lab-Anti-PhishingPolicy applied.
Confirmed	Two URLs were extracted; the displayed threat verdict for each was None.
Confirmed	No attachments were present.
Not established	A malicious URL click, Safe Links block, credential compromise, or endpoint execution.

MITRE ATT&CK context

T1566 — Phishing is the most relevant technique for the simulated email-delivery attempt. This is an analyst mapping of the scenario, not a claim that the test produced a confirmed compromise.

Outcome and lessons learned

Microsoft Defender initially detected and quarantined the simulated $485.75 refund phishing email using user impersonation and advanced filtering. This confirmed that the configured anti-phishing protection functioned as intended during initial message processing.

The email was subsequently released by an authorized administrator as part of the controlled lab exercise. Defender later recorded one allowed click on the embedded refund-verification URL.

This investigation demonstrated the importance of distinguishing initial phishing detection, administrative release actions, and URL-level protection. Passing SPF, DKIM, and DMARC checks does not establish that an email is safe. The available evidence did not establish malicious URL blocking, credential theft, malware execution, or account compromise.

Portfolio skills demonstrated

Phishing triage • Microsoft Defender XDR • Microsoft 365 email security • Quarantine analysis • Email authentication interpretation • URL analysis • Evidence-based SOC reporting • MITRE ATT&CK mapping

Evidence source: screenshots embedded in the original lab report supplied for this case study. All activity described is an authorized security simulation.

Additional Evidence 06 — KQL Detection Validation

Original $485.75 refund phishing email | Microsoft Defender XDR Advanced Hunting

Investigation update — October 14, 2026

Objective

Independently verify the original refund-themed message's detection signals and delivery disposition using the EmailEvents table, scoped to its unique NetworkMessageId. This section excludes the separate Safe Links test message.

Hunting query

```
EmailEvents
| where NetworkMessageId == "83eec52e-9473-4c3d-9f6d-08df29fb4a37"
| project Timestamp, Subject, SenderFromAddress,
        RecipientEmailAddress, ThreatTypes,
        DetectionMethods, EmailAction,
        EmailActionPolicy, DeliveryAction,
        DeliveryLocation, NetworkMessageId
```

Observed results

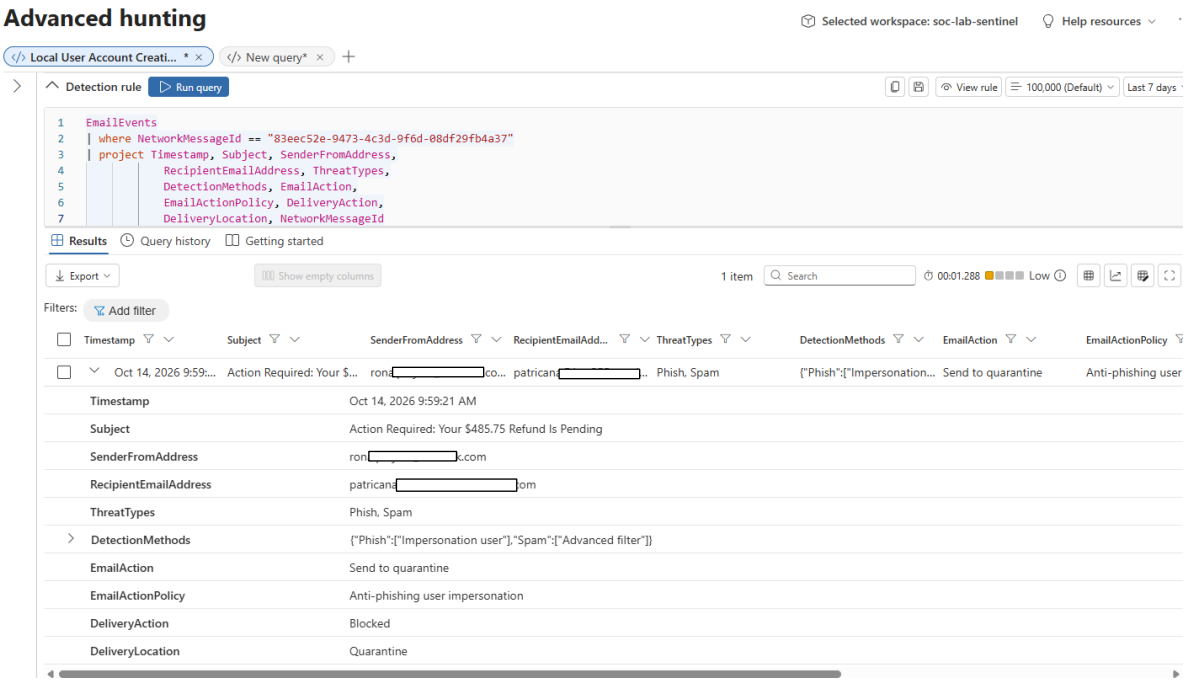
Matched records	1
Subject	Action Required: Your $485.75 Refund Is Pending
Timestamp (displayed)	October 14, 2026, 9:59:21 AM
Sender	ronxxxxxx@[redacted]m
Recipient	patrica@xxxxxxx.[redacted]m
ThreatTypes	Phish, Spam
DetectionMethods	Phish: Impersonation user; Spam: Advanced filter
EmailActionPolicy	Anti-phishing user impersonation
EmailAction	Send to quarantine
DeliveryAction / Location	Blocked / Quarantine

Analyst assessment

The query returned one record for the original refund email. Its detection metadata attributes the phishing classification to user impersonation and the spam classification to advanced filtering. The recorded policy action was to quarantine the message, and the delivery fields confirm it was blocked and placed in quarantine. This corroborates the quarantine screenshots in the original investigation.

Limitations of Evidence 06 considered alone: The EmailEvents query is scoped to one message ID and does not establish whether other related messages exist. Subsequent Evidence 07 and 08 address URL interaction and administrative release; neither demonstrates credential compromise.

Evidence screenshot



Evidence 06 — EmailEvents Advanced Hunting result for the original refund phishing message.

Evidence 07 — URL Click Investigation

Original refund email | Microsoft Defender XDR Advanced Hunting | October 14, 2026

To assess recipient interaction, I searched UrlClickEvents using the original email's NetworkMessageId within a 24-hour time window. The query returned one matching event.

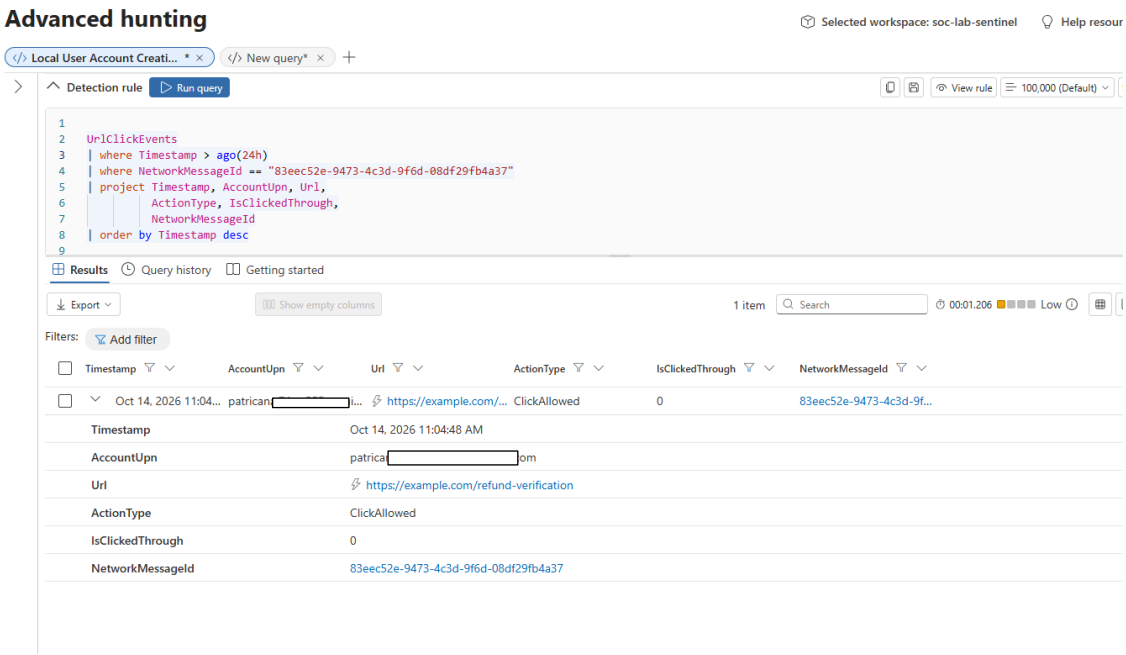
Hunting query

```
UrlClickEvents
| where Timestamp > ago(24h)
| where NetworkMessageId == "83eec52e-9473-4c3d-9f6d-08df29fb4a37"
| project Timestamp, AccountUpn, Url, ActionType, IsClickedThrough, NetworkMessageId
| order by Timestamp desc
```

Recorded event time	October 14, 2026, 11:04:48 AM (portal display)
Recipient	patrican[redacted]om
URL	https://example.com/refund-verification
ActionType	ClickAllowed
IsClickedThrough	0
Correlation	NetworkMessageId matches the original refund email

Analyst interpretation

Defender recorded an allowed URL-click event associated with the original phishing simulation. IsClickedThrough = 0 does not indicate a user bypass of a Safe Links warning. This is evidence of a recorded click, not of malicious URL blocking, credential entry, malware execution, or account compromise.



Evidence 07 — UrlClickEvents result, filtered to the original refund message within 24 hours.

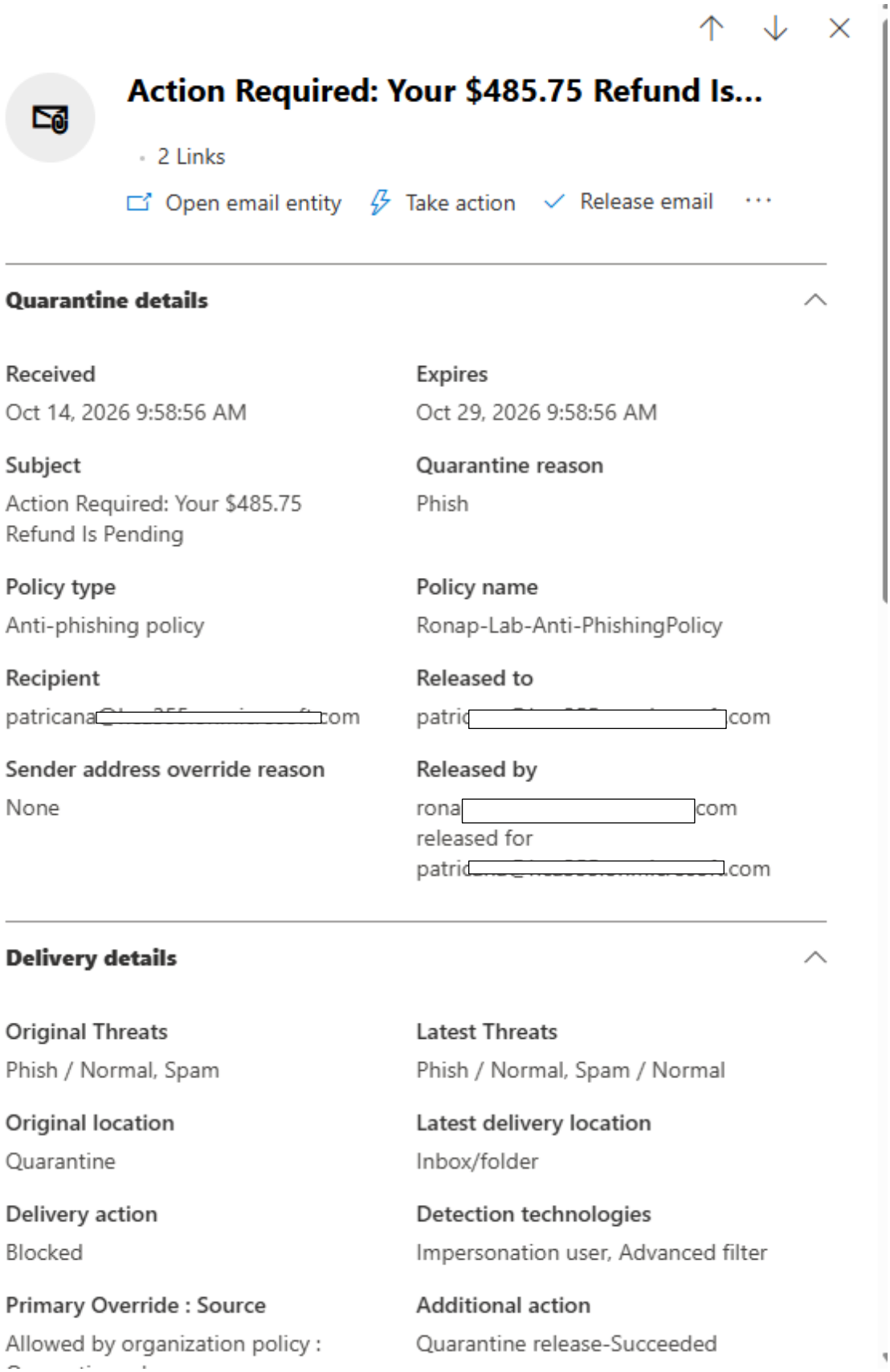
Evidence 08 — Quarantine Release Verification

I revisited the original message in Microsoft Defender Quarantine to establish its post-quarantine disposition.

Original quarantine reason	Phish
Policy	Ronap-Lab-Anti-PhishingPolicy
Original delivery action / location	Blocked / Quarantine
Released by	ronap@xxxxxx.[redacted].com
Released to	patricana@xxxxx[redacted]com
Latest delivery location	Inbox/folder
Additional action	Quarantine release-Succeeded
Release timestamp	Not shown in the captured evidence

Analyst interpretation

The release was a deliberate, authorized action in the controlled lab. It explains a plausible route for later recipient interaction without contradicting the original blocked/quarantined EmailEvents result. The screenshot does not establish the exact release time, so the chronological order of release versus click is not independently verified from timestamps.



Evidence 08 — Original refund email: successful quarantine release and latest inbox/folder delivery location.

Final Investigation Timeline & Disposition

October 14, 2026, 9:58:56 AM	**Original refund email received; quarantine record displays Phish classification.**
October 14, 2026, 9:59:21 AM	EmailEvents records Phish/Spam, impersonation user / advanced filter, and Blocked → Quarantine.
Release time not displayed	Authorized administrator released original message; quarantine view confirms success and Inbox/folder as latest location.
October 14, 2026, 11:04:48 AM	UrlClickEvents records ClickAllowed for the refund-verification URL; IsClickedThrough = 0.

Final analyst assessment

The controlled refund-themed email was initially detected and quarantined by Microsoft Defender anti-phishing protections. KQL correlation independently verified the detection signals and initial delivery decision. An authorized administrator later released the message to the intended lab mailbox, and a matching allowed URL-click event was recorded. The initial filtering control functioned as observed; the later inbox delivery was an administrative lab action, not evidence that the original quarantine failed. The supplied evidence does not establish credential theft, malware execution, or account compromise.

Disposition

Authorized security simulation — expected test activity. Detection outcome: initial quarantine successful. Administrative release: successful. URL telemetry: one allowed click recorded. No confirmed compromise in the supplied evidence; absence of compromise is not independently proven.

Scope and limitations

This case is limited to the original $485.75 refund message and its unique NetworkMessageId. A single matching EmailEvents record does not establish that no related messages were sent. The exact release timestamp and a complete downstream impact assessment are not available in the screenshots. The separate Safe Links test message is excluded from this case study.

Portfolio skills demonstrated

Phishing triage; email authentication interpretation; Microsoft Defender XDR Advanced Hunting; KQL filtering and message-ID correlation; detection-method analysis; quarantine and administrative-release verification; Safe Links click-telemetry interpretation; evidence handling; and defensible SOC reporting.

Production-response note

In a real phishing investigation, analysts should not release suspicious mail simply to test its links. Recommended next actions would include scoping similar messages, checking recipient interaction, and assessing account impact using authorized response procedures. These are recommendations, not additional actions claimed in this lab.